ON Earth Life

MAGZAZINE

#1

TOPICS

atomic level

fun facts

nervous system - how to use it

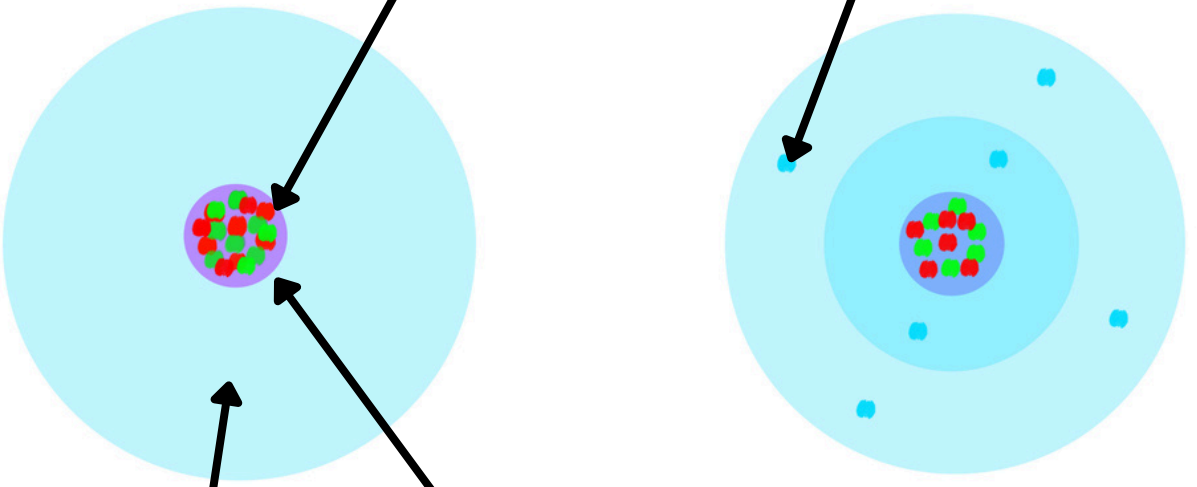
A reality is layered. And what both Einstein and Stephen Hawking proved - there is no easy and quick way to understand it.
But layer by layer, topic by topic, it starts to build a somewhat sensible picture behind your mind’s eye.

First layer, known by many, is the atomic structure of reality. It really is not exactly what it seems like, but through the publications we will dig deeper into its nature. And it's also not the first layer but we will be starting the explanations from here.

Don't get discouraged, because this type of knowledge seems absolutely abstract, but the more puzzles of it you get, the more sense it starts to make. And once it 'clicks', you start to understand the hidden yet undeniable and crucial logic, which rules our world and therefore you and others.

You can treat this level as facts. Don't be later surprised however, that broadening this topic may make this truth seem a little outdated or definitely incomplete.

Most of the ordinary matter around us is made of absurdly huge amounts of connected (or not) atoms. Every atom in most cases is made of 3 other particles - protons, neutrons and electrons, arranged in a specific way.



singular carbon atom

- 6 protons
- 6 neutrons
- 6 electrons

area where electrons can be found

Protons and neutrons form a dense nucleus in the center, while electrons are found around it.

(about other crucial particles like light particles in another magazine)

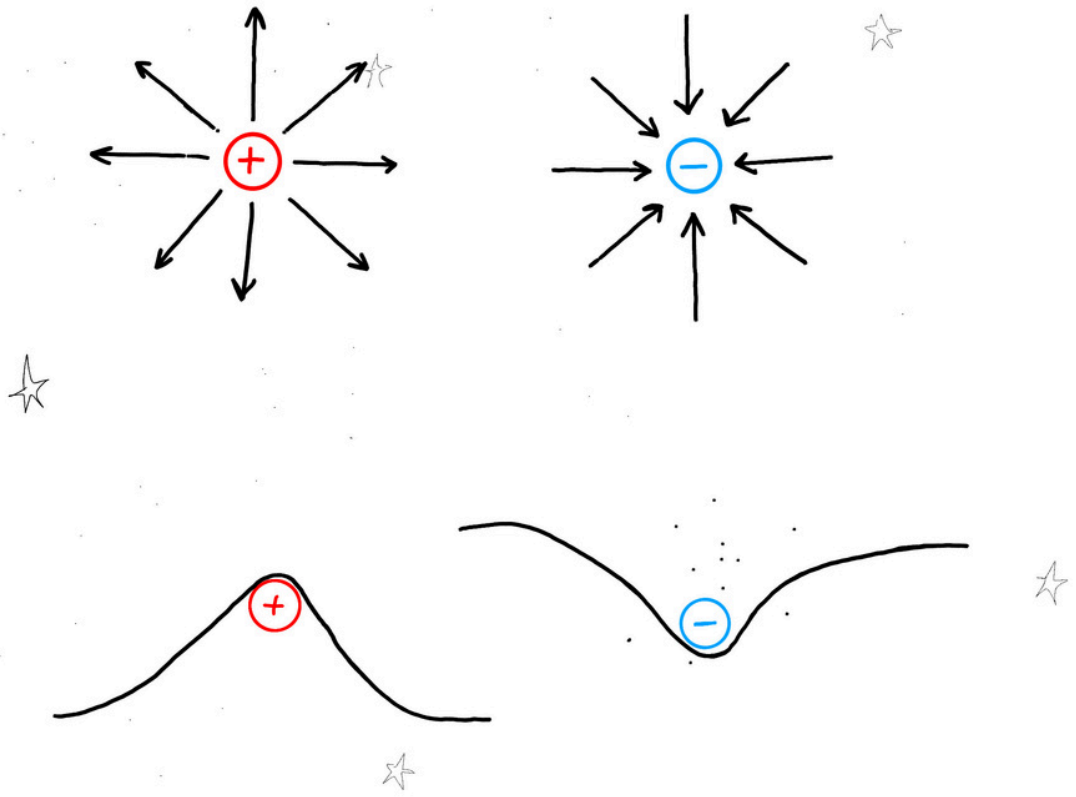
EVERY particle has energy, not every particle has an electric charge

And definitions in physics aren’t always straight forward.



Energy describes the capacity of an object to do a certain amount of work - it describes how much work can an object do.

From what we know about electric charge it can be positive, negative, or zero. The same charges repel each other and different charges attract. They also react differently with the field (as shown below).



However, except for these characteristics, we still have no idea why the differences between these charges exist. Because except for behaving differently, they are built exactly the same.

SO THE FIRST LAYER ARE ATOMS, from which you are entirely built

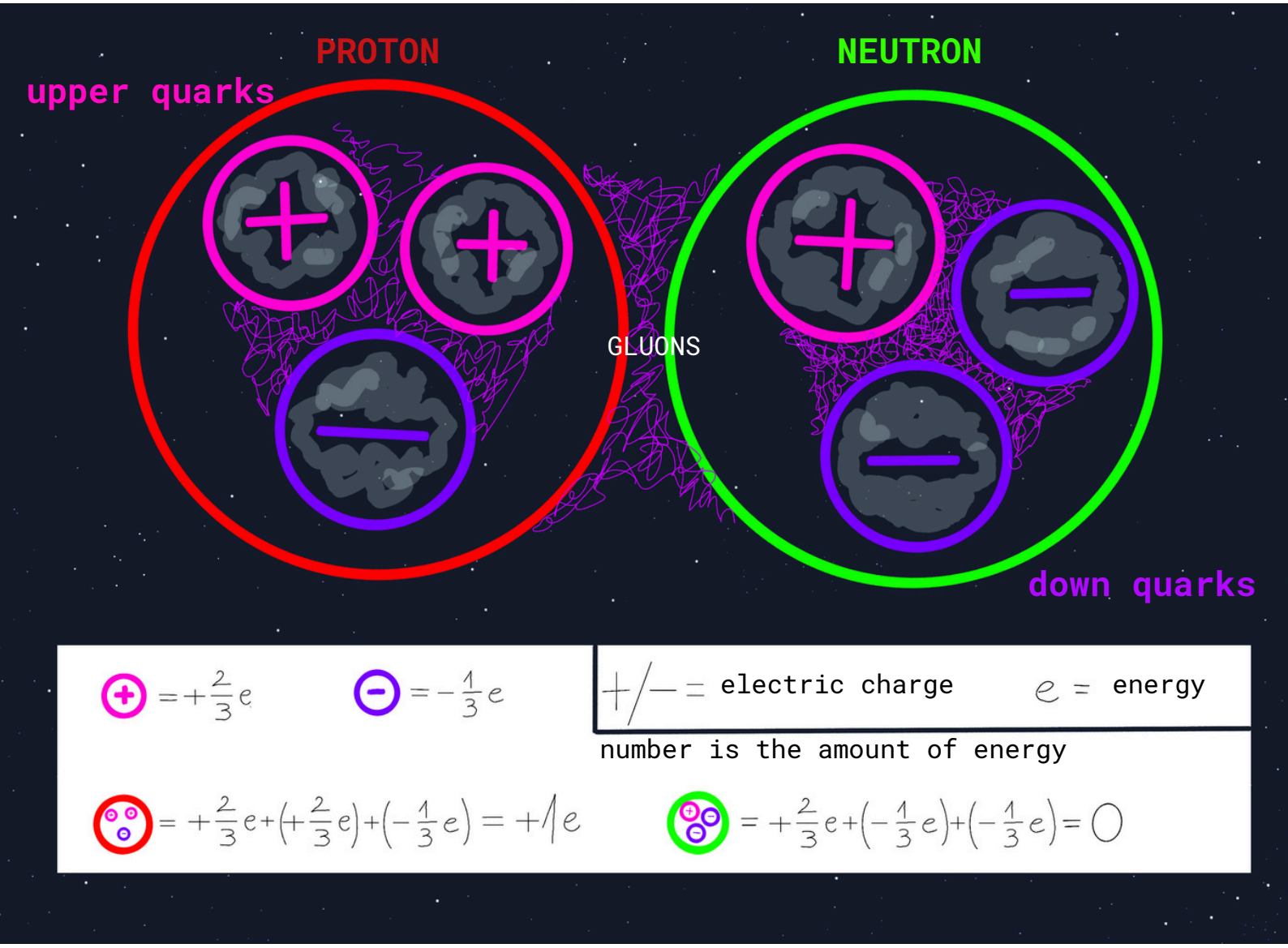
EACH ATOM (usually) IS BUILT OF 3 OTHER PARTICLES protons, neutrons and electrons

EVERY ONE OF THESE PARTICLES HAS ENERGY
***NOT EVERY OF THESE PARTICLES HAVE AN ELECTRIC CHARGE**

***SOME OF THIS PARTICLES ARE MADE OF EVEN SMALLER PARTCLES**

*explanation in next pages

Inside a proton and a neutron, there are other, smaller particles - **quarks** that have electric charges and that are held together by the strong interaction. This interaction is mediated by other particles called **gluons**. They are very much needed, because as mentioned inside both proton and neutron there are particles - quarks - with the same electric charge, therefore, they constantly repel one another.



A proton has a net positive electric charge, while a neutron has no net electric charge.

The strong interaction (provided by gluons) is much more important than the electromagnetic repulsion of the similarly charged quarks at the extremely small distances inside a proton or neutron. So it keeps the quarks bound together.

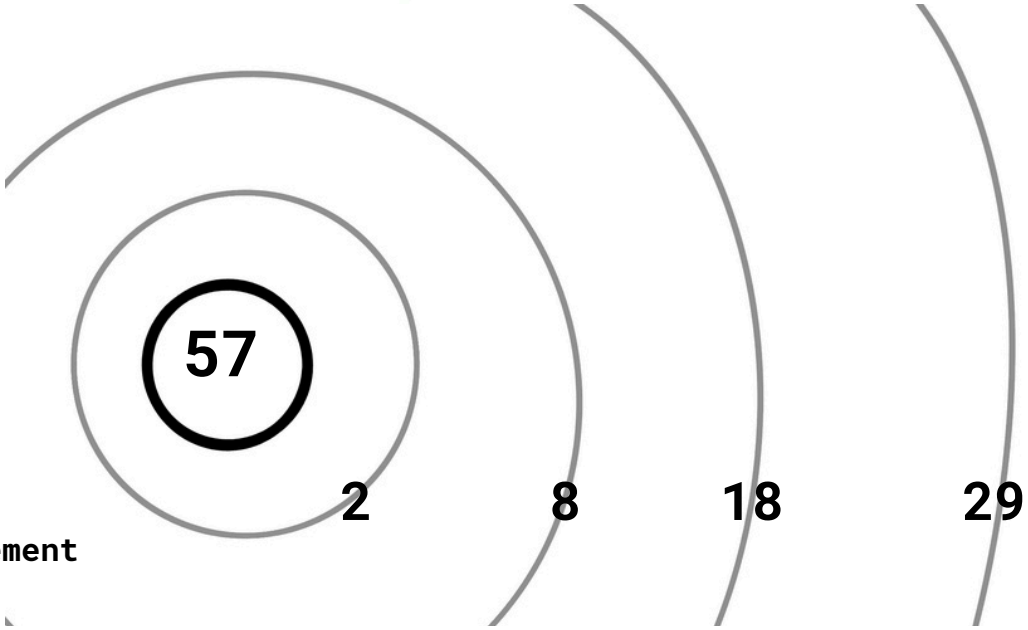
Gluons also provide the force to keep protons and neutrons together in the form of a nucleus which is very much needed since similarly charged protons constantly repel one another.
Protons and neutrons have one fixed amount of energy.

In an atom, electrons carry negative electric charge and, as far as we know, they are not made of any other, smaller particles*. But electrons don't have just one fixed amount of energy. They can exist in certain, different but fixed energy states. You can think of these states as different energy levels. And each level can hold only a fixed and limited number of electrons.

*electrons and quarks (and some others) are fundamental particles.

An electron in one of these energy levels does not sit in one particular spot, like a tiny ball sitting at a specific point. Quantum physics tells us that, before we measure its position, we cannot say exactly where the electron is. Instead, we can describe the probability of finding it in different places around the nucleus.

This popular explanation from school is widely misunderstood:



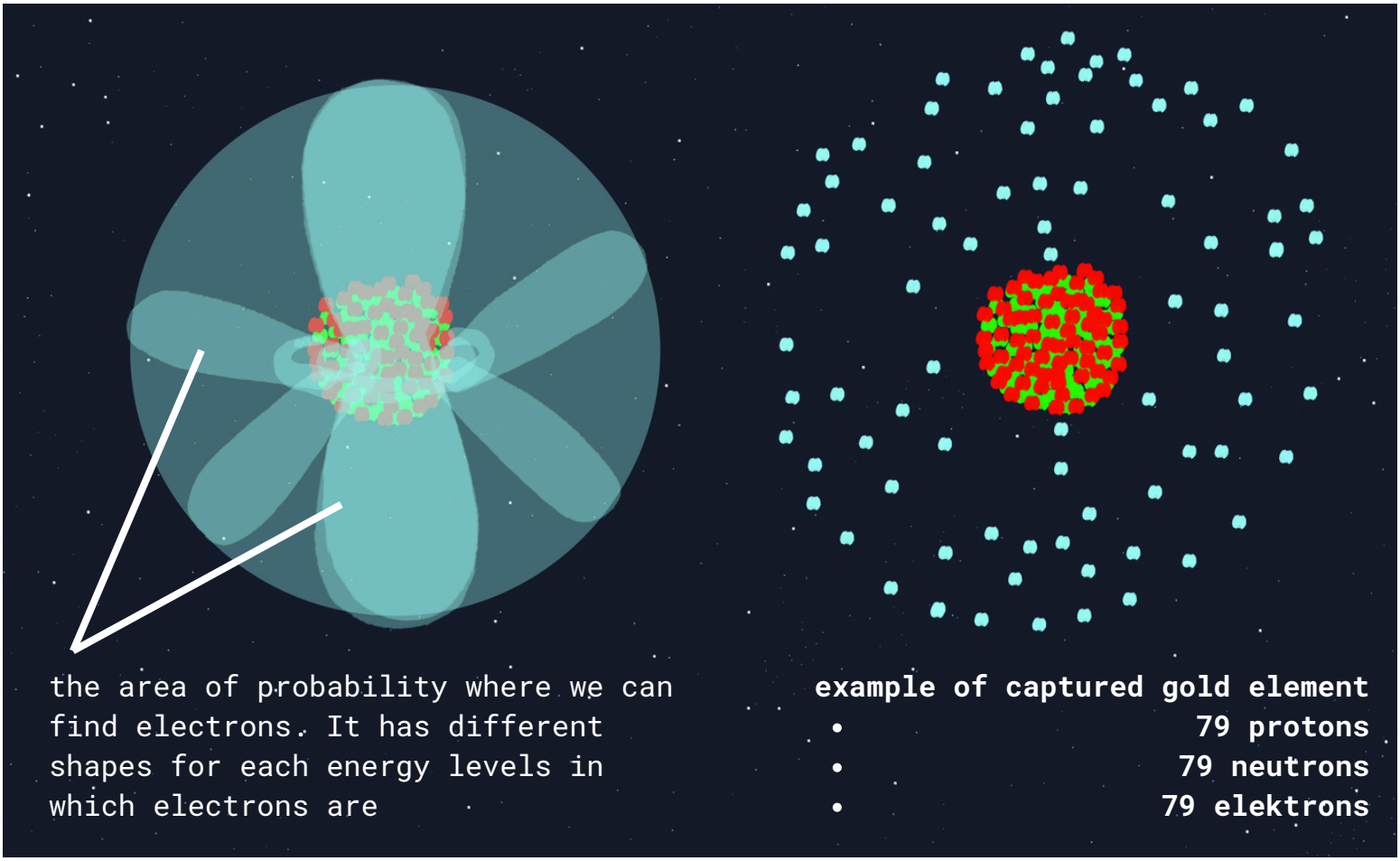
lantan element

These are not layers explaining where and how many electrons are there. This only explains the energy levels and how many electrons can be in each energy level.

And science can’t describe yet how exactly this uncertainty influences the reality.

the fact that one of the indispensable parts of the matter you are made of can be found in various positions, might mean that indeed we have a free will and what we see might differ

We can tell how many electrons are in an atom and therefore how many electrons are in each energy level (but still… we cannot define their exact location until we somewhat capture them).



the area of probability where we can find electrons. It has different shapes for each energy levels in which electrons are

example of captured gold element

- **79 protons**
- **79 neutrons**
- **79 elektrons**

We differentiate atoms with different amounts of protons in the nucleus into elements.

All elements which we know are described in Mendeleev’s periodic table.

The number of neutrons is always the same as the number of protons. The number of electrons can differ, but usually in atoms basic state it is also the same as the number of protons.

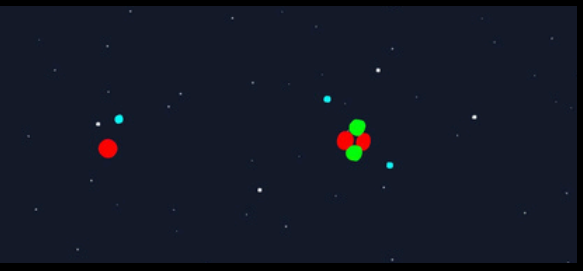
Atoms connect together by ‘connecting’ the outer part - electrons.
After connecting, particular electrons from different atoms occupy quantum states (energy levels) that extend over two or more atoms that connect together.

So they don’t connect in a way like we usually understand it - they don’t stick together. Rather they just share electrons - some part of themselves with one another.

some

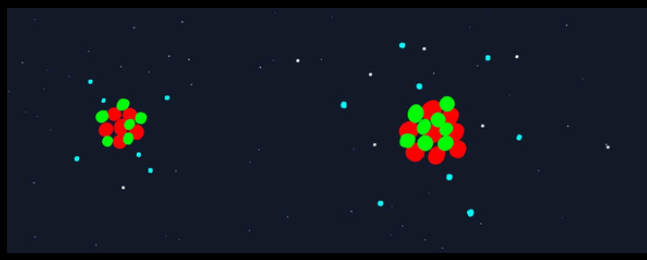
FUN FACTS

Most atoms that build our bodies were formed in stars.
The first stars formed from enormous amounts of light elements (with not many protons in their nuclei), mainly hydrogen and helium.



A hydrogen atom normally contains one proton and one electron, while a helium atom contains two protons, two neutrons and two electrons.

In the very central part of a star, there was the biggest accumulation of these elements. As gravity pulled this material together, the pressure and temperature in the center of the star increased enormously. This, in turn, created conditions in which hydrogen and helium nuclei could undergo nuclear fusion, combining lighter nuclei into heavier ones. Among the newly created elements were carbon and oxygen,



which are crucial for the structure of our bodies.
In short, then a star exploded and the newly created elements were released into space.
And on our planet they were first found in water. And in this rich in elements water another level of our bodies (and every other living creature’s) structure formed itself - cells.

Cells are made of atoms arranged in a system that can obtain and transform energy from the environment - which is from other particles.

Some organisms are made of just a single cell – bacterias

Like mentioned cells primarily developed themselves in water
And to this day, every cell can actively live ONLY in a water-rich environment.

Together, cells can build bodies of various living creatures – trees, humans, lions, mushrooms, birds, flowers, etc.

So basically, we (and everything else that lives) are made of enormous amounts of tiny connected water organisms, which connected together can create a species that can live on land.

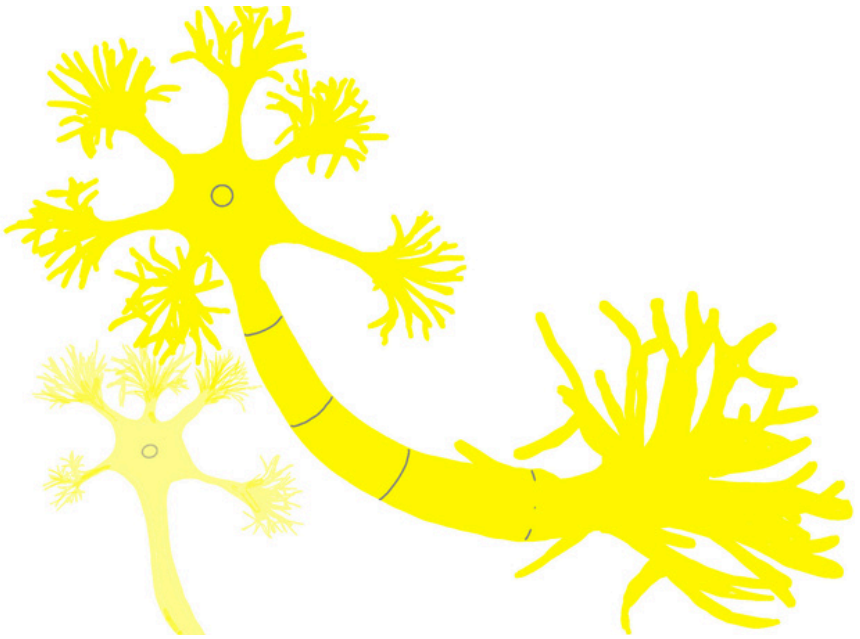
Nevertheless, water is still crucial for energy production and therefore for the survival of ALL Earth creatures.

And the energy in your cells is the same energy as there is in a lightbulb.

NERVOUS SYSTEM

the structure in human bodies that determines how we percieve / feel the outside world and how we interact with it

The nervous system is built from cells called neurons. They have a shape that is, in some ways, similar to a tree.



And in some way they even work in a similar way.

These types of cells have two main functions in the human body:

1. They receive and transmit information from the outside world to the brain (like what, where and how interacted or may interact with the body)
2. They receive and send information from the brain to effector organs, such as muscles to make an intentional movement.

The brain is made of highly condensed neurons, which are connected together in very specific and complicated ways. This precision in their connections allows them to produce reactions that are appropriate to whatever circumstances are happening outside or to whatever we want to happen.

The nervous system is responsible for our reactions to the outside world and for our actions in it.

We can, in some way, **control it through our thoughts**.

For example, imagine you are in a dark forest. There are probably two ways you can approach this situation. You can be scared or you can be calm, and it may depend on how you think about the situation—whether you think there is a threat awaiting you, or whether you think that you already know the forest and that there is probably nothing to be afraid of, even though it is dark.

An aroused nervous system can appear as

- increased blood pressure,
- overthinking,
- tense muscles,
- increased sweating,
- difficulty falling asleep.

A calm nervous system appears as

- regular blood pressure,
- relaxed muscles,
- ease of concentration,
- general calmness.

Some functions of the nervous system we can control, while others we cannot.

We can control the movement of the body that we want to initiate.

We cannot control the fact that information reaches our brain, for example pain sensations (without pharmacological intervention)

We can, however, **always regulate** its **activity** whether it’s aroused or calm .

A state of activity defines how we react to whatever is happening in the world –whether we react to the environment with calmness and peace or fear and with a sense of threat.

The first way to change from an aroused to a calm nervous system is to regulate it through breath.
A long exhale and a shorter inhale, repeated many times, gives the body information that the organism does not need a lot of oxygen because there is no need to support the muscles, as so there is no need to use them. This signals that there is no threat that you need to run away from or to fight with. Because when your nervous system is aroused your brain subconsciously reads the situation as potentially life threatening and so it puts the body into a state of steadiness and preparation rather than calmness.

The second way to change from an aroused to a calm nervous system is to observe your environment and your inner state in an emotionless way. Notice exactly what is around you and what situation you are in, and try not to connect it with any emotion. Then notice what emotions are accompanying you. Acknowledge them, but do not let them control your actions.
Thanks to this, your brain changes into a state of clarity in which there is nothing it has to interpret anymore and nothing is hidden from it, so it is focused on the current situation with clarity.

Now imagine a situation

someone is screaming at you. They have an aroused nervous system.

Our brains have adapted to social cues, so when the brain sees that someone is upset, it will most likely also shift into a more aroused state. It is like that because, in the past, the nervous reaction of someone else could inform us about a danger and about the need to switch to a more aroused state of the nervous system in order to prepare to escape or fight. Luckily, nowadays, information about danger can be communicated verbally, and someone else's aroused nervous system does not always inform us about something that may also affect us.

So in order to keep calm during a situation like that, you can make a simple observation:

I see that this person is screaming at me.
I feel that there is anger growing in me.
What is this person trying to tell me?
(Now you are listening to the person and trying to make sense of what they are saying).
Okay, so this person is speaking about X, Y, Z.
What do I think about X, Y, Z? Or what do I think about the current situation in which we both are?
Now how do I prefer to communicate?
How do I usually communicate when my nervous system is in a calm state? And therefore, when it is in a state where it can form and communicate thoughtful and logical responses rather than passing emotions.

Now take a breath, make a longer exhale, and try to interact with this situation from a place of calmness and reason rather than fear or anger.

Why is it important?

Because by training your nervous system, you can react to situations in a logical way rather than through impulses, which are often misguided.

Generally, our brain is a very plastic tool. We can transform it and shape it as we want it to be.
A brain that is not trained has trouble logically connecting facts, making quick and accurate deductions, staying calm when facing changes, as well as adapting to changes.

and changes are an inevitable part of life. So it's better to react to them with a calm rather than fearful mind.

An untrained brain works through impulses rather than free will. **If you want to maintain peace of mind and independent, uninfluenced thinking**, then training your brain is an effective way to do that.

A trained brain is also easier to regulate. Training your brain means enhancing its capacity and strengthening neural connections, which allows you to form quick and accurate conclusions and develop an effortless awareness of yourself and your surroundings.

The development and strengthening of these connections is supported by:

- learning new skills and information,
- reading,
- regular and interesting physical activity,
- good-quality sleep (more about sleep hygiene and the idea of dreams will be found in another magazine).
- an appropriate diet (rich in: vegetables, fruits, fatty fish, olive oil, nuts, natural yogurt, kefir, rice, grains, and whole-grain bread. Low in: highly processed foods– foods whose ingredients you can hardly recognize and understand –as well as excessive amounts of sugar, salt, and alcohol).

Alcohol, alongside a lack of social relationships, is one of the important risk factors associated with Alzheimer’s disease and dementia.

If you really want to see the results, it is more about making these changes a part of life rather than doing them from time to time. But it's always good to start somewhere. And an enhanced self awareness and brain abilities appear in way more visible ways than you can imagine.

The way you think shapes your feeling of life,
because most situations can be perceived through many different perspectives.
You choose which one you live by.
And the ability to choose is something to be trained and tested,
since there are endless options.

Through every part we will brighten the understanding of reality.

Continously, by diving deeper in it in first articles





peace, Aeria Tellar



